ENGIFLOW AI | SAMPLE ELECTRICAL ENGINEERING SPECIFICATION

FICTIONAL TRAINING DOCUMENT - NOT FOR DESIGN, PROCUREMENT, OR CONSTRUCTION

Project: Example Data Center Power Infrastructure | Revision: A | October 2026

1. Project overview

This fictional example describes a 2 MW data center electrical distribution concept. The purpose is to test automated extraction of engineering document requirements and traceable source citations.

2. Electrical distribution and voltage

The conceptual distribution architecture shall include an **800 VDC** bus for selected computing loads. Incoming utility service is assumed at **13.8 kVAC**, three-phase, stepped down for facility distribution. The AC distribution system operates at **60 Hz**. AC/DC conversion equipment shall be identified on the electrical drawings.

3. Power capacity and loading

The nominal electrical design capacity is **2 MW**. The planning document assumes a **2,500 kVA** main transformer and an illustrative peak IT demand of **1.6 MW**. Capacity ratings are conceptual and require engineering validation.

4. Energy storage

A preliminary battery energy storage system (**BESS**) with usable energy capacity of **500 kWh** is in scope. The energy storage system interfaces with the DC bus through a bidirectional converter. Battery protection and emergency shutdown requirements must be addressed during detailed design.

5. Protection, safety and grounding

Provide appropriately rated **circuit breakers**, overcurrent protection, grounding/bonding details, and an emergency power-off (EPO) strategy. A short-circuit and arc-flash assessment shall be completed by qualified engineering professionals. The illustrative main breaker rating is **3,000 A**, subject to coordination study.

6. Drawings and single-line diagram

The engineering package shall include a **single-line diagram (SLD)**, drawing reference **E-001 Rev A**, showing utility service, transformer, AC switchgear, rectifiers, 800 VDC bus, BESS, and principal loads. Include a drawing revision register for future design updates.

7. Grid interconnection and utility

Coordinate the **utility interconnection** at the point of common coupling (PCC). A grid impact / interconnection review is required before finalizing capacity, metering arrangements, and protection interfaces. The specific utility and interconnection approval status are **not specified** in this sample.

8. Testing and commissioning

The commissioning plan shall include **factory acceptance testing (FAT)**, **site acceptance testing (SAT)**, insulation checks, protective relay functional checks, and verification of the EPO sequence. Test procedures and acceptance criteria require approval before field execution.

9. Open items for engineering review

- Confirm actual utility provider and interconnection approval timeline.
- Finalize electrical load study, redundancy requirements, and fault-duty calculations.
- Confirm battery chemistry, ventilation, fire-protection approach, and code applicability.
- Validate transformer / breaker ratings and the 800 VDC equipment ecosystem.

Notice: This document is intentionally fictional. Values are test fixtures for EngiFlow AI and are not recommendations or an approved design.